Let us express the above pattern of shapes in the form of a table.

Let x denote the number of steps and y denote the total number of circular rings.

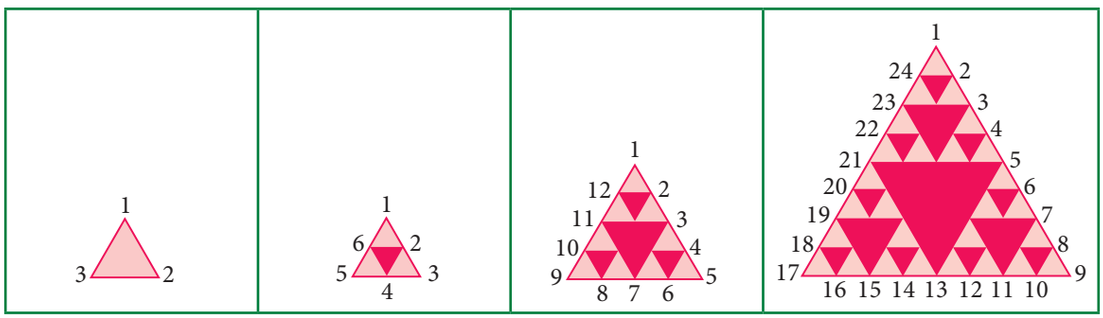
Number of steps (x)	1	2	3	4	…
Number of circular rings (y)	1	3	5	7	…

Here let us find the relationship between the number of steps and number of circular rings.

Hence the relationship between x and y can be generalised as $y = 2x - 1$

Situation 2

Consider an equilateral triangle of any size  . Mark the midpoints of sides of a triangle and join them to form four equilateral triangles  . Proceed in the same way to get the shapes following a pattern as shown below.



Let us express this repeated process in the form of a table.

Here let us find the relationship between the number of steps (x) and total number of vertices (y).

Number of steps (x)	1	2	3	4	…
Number of vertices (y)	$3\times2^0=3$	$3\times2^1=6$	$3\times2^2=12$	$3\times2^3=24$	…

Hence, the relationship between x and y can be generalised as $y = 3\times\left(2^{x-1}\right)$.

From the above situations, you could have understood that even if an action is repeatedly done, the properties of the shape do not change.

Example 5.1

In the following figures, polygons are formed by increasing the number of sides using matchsticks as given below.

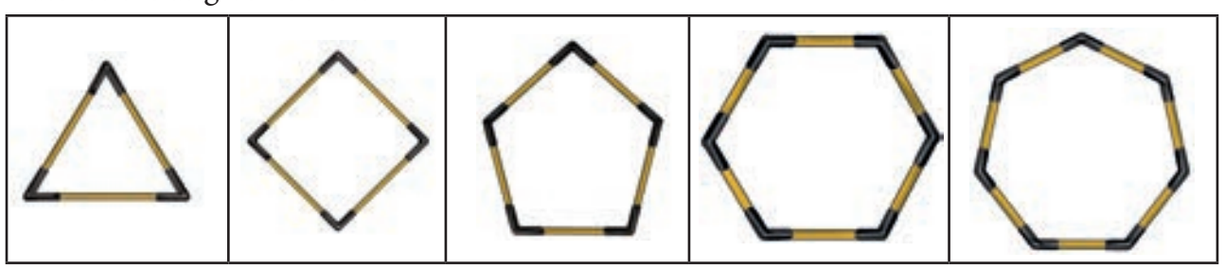


Fig. 5.2

Find the number of sticks required to form the next three shapes by tabulation and generalisation.

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